

Education

Military Institute of Science and Technology (MIST) — *M.Sc. Engg in Computer Science and Engineering* Apr 2026 – Present

Southeast University — *BSc in Computer Science and Engineering* Oct 2021 – Oct 2025
CGPA: 3.87/4.0

Relevant Coursework: Artificial Intelligence, Image Processing, Data Mining, Research Methodology

Experience

Research Assistant — *Islamic University of Technology (IUT), OIC* Gazipur, BD
Nov 2025 – Present

- Engineered a Federated Learning framework for fintech data with a novel aggregation technique for non-IID settings.
- Pioneered hyperbolic Graph Neural Networks for blockchain fraud detection and causal inference methods for high-dimensional auditing.
- Authored five manuscripts accepted in Q1 journals on decentralized AI, graph-based fraud detection.

Research Assistant — *CAIR, Southeast University* Dhaka, BD
Jan 2025 – Oct 2025

- Designed hybrid transformer-based architectures for medical image segmentation and classification, fusing convolutional and attention mechanisms to improve diagnostic accuracy on MRI and X-ray datasets.
- Integrated explainable AI methods (SHAP, LIME) into clinical decision-support pipelines, enabling interpretable model outputs for high-stakes diagnostic settings.
- Authored five papers: three accepted at peer-reviewed conferences, two under review in Q1 journals on medical AI and interpretable deep learning.

Selected Publications

MangoLeafNet-XAI: An Attention-Enhanced Deep Learning Architecture for Accurate and Interpretable Mango Leaf Disease Classification [Paper]

Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Farzan Majeed Noori, Md Zia Uddin, Abdul Kadar Muhammad Masum

Frontiers in Plant Science [Q1] [🔗](#)

Integrating Domain-Specific Convolutional Features with Swin Transformers for Transparent Cervical Cancer Diagnostics [Paper]

Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Md Fokrul Islam KHAN, Md Zia Uddin, Ulysse Teller Masao Côté-Allard, Abdul Kadar Muhammad Masum

Frontiers in Oncology [Q2] [🔗](#)

FedMal-XAI: An Explainable Federated Vision Transformer Leveraging Knowledge Distillation for Privacy-Preserving Malaria Detection [Paper]

Md. Tofael Ahmed Bhuiyan, **Md. Abdur Rahman**, Md Fokrul Islam Khan, Farzan M. Noori, Abdul Kadar Muhammad Masum

Frontiers in Public Health [Q1] [🔗](#)

DFU-GCNet: A Global Context-Enhanced Inception Network for Robust and Interpretable Diabetic Foot Ulcer Classification [Paper]

Md. Tofael Ahmed Bhuiyan, **Md. Abdur Rahman**, Farzan M. Noori, Abdul Kadar Muhammad Masum

Frontiers in Digital Health [Q1] [🔗](#)

Privacy-Preserving ESG Risk Prediction via Federated Hybrid Neural-Boosting Ensembles [Paper]

Abdul Kadar Muhammad Masum, Md. Abul Kalam Azad, **Md. Abdur Rahman**, Md. Tofael Ahmed Bhuiyan, Ramona Birău, Virgil Popescu

IEEE Access [Q1] [🔗](#)

Predicting Corporate Bankruptcy Through a Scalable and Interpretable Knowledge Distillation Framework [Paper]

Md. Abul Kalam Azad, Abdul Kadar Muhammad Masum, **Md. Abdur Rahman**, Md. Tofael Ahmed Bhuiyan, Thanh Lam Nguyen, Hasan Dincer, Serhat Yuksel

Discover Artificial Intelligence [Q1] [🔗](#)

- EcoStack-Pro: An Adaptive Federated Learning Framework for Interpretable ESG Auditing Across Heterogeneous Industrial Sectors** [Paper]
Md. Abul Kalam Azad, Abdul Kadar Muhammad Masum, *Md. Abdur Rahman*, Md. Tofael Ahmed Bhuiyan, Farzan M. Noori, Md Zia Uddin
[Frontiers in Artificial Intelligence \[Q1\]](#) [↗](#)
- Bridging Data Silos in Corporate Governance: A Hierarchical Stacking Ensemble with Federated Dynamic Aggregation** [Paper]
Abdul Kadar Muhammad Masum, Md. Abul Kalam Azad, Md. Tofael Ahmed Bhuiyan, *Md. Abdur Rahman*
[Applied Computational Intelligence and Soft Computing \[Q1\]](#) [↗](#)
- Beyond Short-Term Noise: Robust Long-Horizon Bankruptcy Prediction Using Gated Dual-Branch Transformers** [Paper]
Rezwanul Huque Khan, Md. Abul Kalam Azad, Md. Tofael Ahmed Bhuiyan, *Md. Abdur Rahman*, Faisal Md. Abdur Rahman, Abdul Kadar Muhammad Masum
[Applied Computational Intelligence and Soft Computing \[Q1\]](#) [↗](#)
- Do LLMs Distinguish Between Halal and Haram? Benchmarking Islamic Cultural Alignment in General vs. Arabic-Centric SLMs** [Paper]
Md Abdur Rahman, Md Tofael Ahmed Bhuiyan, Abdul Kadar Muhammad Masum
[6th Muslims in ML Workshop, ICML 2026](#) [↗](#)
- Tokenizer Transplantation: Mitigating Autoregressive Collapse in Edge-Efficient Bengali ASR** [Paper]
Sanjid Hasan, *Md Abdur Rahman*
[6th Muslims in ML Workshop, ICML 2026](#) [↗](#)
- SFL-MTKD++: Split ViT Federated Learning with Adaptive Multi-Teacher Distillation for Interpretable Malaria Diagnosis** [Paper]
Md Tofael Ahmed Bhuiyan, *Md Abdur Rahman*, Abdul Kadar Muhammad Masum
[6th Muslims in ML Workshop, ICML 2026](#) [↗](#)
- AlzFed-XAI: High-Fidelity Interpretable Alzheimer’s Diagnosis with Privacy-Preserving Federated Learning** [Paper]
Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Abdul Kadar Muhammad Masum
[5th Muslims in ML Workshop, NeurIPS 2025](#) [↗](#)
- ChakmaBridge: A Five-Way Parallel Corpus for Navigating the Script Divide in an Endangered Language** [Paper]
Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Abdul Kadar Muhammad Masum
[2nd Bangla LP Workshop, AACL-IJCNLP 2025](#) [↗](#)
- BhasaBodh: Bridging Bangla Dialects and Romanized Forms through Machine Translation** [Paper]
Md. Tofael Ahmed Bhuiyan, *Md. Abdur Rahman*, Abdul Kadar Muhammad Masum
[2nd Bangla LP Workshop, AACL-IJCNLP 2025](#) [↗](#)
- Detecting Manipulation in Ukrainian Telegram: A Transformer-Based Approach to Technique Classification and Span Identification** [Paper]
Md. Abdur Rahman, Md Ashiqur Rahman
[4th Ukrainian NLP Workshop \(UNLP 2025\), ACL](#) [↗](#)
- FAKD-XAI: Feature-Aligned Knowledge Distillation with Explainable AI for Efficient Brain Tumor Classification** [Paper]
Md. Abdur Rahman, Sabik Aftahee, Md Ashiqur Rahman, Lamim Zakir Pronay
[ICDSAIA 2025](#) [↗](#)
- X-ArsenicDistill: An Efficient and Interpretable Framework for Arsenicosis Classification via Knowledge Distillation** [Paper]
Md. Abdur Rahman, Md. Tofael Ahmed Bhuiyan, Mohammad Mahmudur Rahman, Iffat Sania Hossain, Abu Kowshir Bitto, Prof. Dr. Abdul Kadar Muhammad Masum
[28th ICCIT 2025](#) [↗](#)
- DeepSea-Net: A YOLO Based Framework for Real-Time Detection and Classification of Underwater Plastic Pollution** [Paper]
Progga Parmita Roy, Sabbir Sarder Munna, Md. Sazed Ul Karim, *Md. Abdur Rahman*, Md. Tofael Ahmed Bhuiyan, Md. Joseb Siddique Siam
[IEEE STI 2025](#) [↗](#)

Under Review Publications

1. **HA-DynAU-Net: A Hybrid Attention Dynamic Aware U-Net for Precise Skin Lesion Segmentation**
[1st Author]
Submitted to *IEEE Open Journal of the Computer Society*
2. **BEACON: A Novel Cross-Modal Architecture for High-Precision and Explainable Brain Tumor Classification Using Medical-Specific Attention**
[1st Author]
Submitted to *PLOS ONE*
3. **HybridTransUNet: Fusing Vision Transformers with Attentive U-Net for Accurate and Interpretable Brain Tumor Segmentation**
[1st Author]
Submitted to *Engineering Reports*
4. **Explainable Artificial Intelligence and Deep Learning for Robust Multi-Class Brain Tumor Classification in Medical Imaging**
[2nd Author]
Submitted to *Scientific Reports*

Honors & Professional Activities

- **3rd Position (Poster Presentation)**, Student Research Day 2025 [Certificate]
- **Champion**, Fragments of Feeling: A Contextual Emotion Challenge (2025) [GitHub]
- **1st Runners-up**, CalorieQuest: Predicting Energy Burn with Data (2025) [GitHub]
- **2nd Runners-up**, DU AI Challenge (2025) [GitHub]
- **2nd Runners-up**, REACT 2026 Datathon, hosted by IEEE SEU Student Branch [GitHub]
- **Finalist (Top 10)**, CUET ETE Datathon 2025 [GitHub]
- **Research Executive**, IEEE Computer Society Bangladesh Chapter Graduate Student Member Forum
- **Program Committee**, 21st BEA Workshop co-located with **ACL 2026**

Experience with Research Grant

Assisted in proposal preparation for the following research grants:

- **Closing the Employability Gap: A Machine Learning–Driven Framework to Predict and Improve Student Placement Outcomes at University** August 2026
Co-PI, IRT Research Grant, Southeast University
Funded amount: \$4,000
- **Bridging the Gap: A Predictive Analytics Framework for Enhancing Engineering Student Employability and Placement Outcomes in Bangladesh** September 2026
Team Member, Bangladesh Accreditation Council (BAC) Research Grant
Funded amount: \$3,000

Professional Certification

- **AWS Certified Cloud Practitioner**, Amazon Web Services Dec 2025 – Dec 2028
- **GitHub Foundations Certification**, GitHub Nov 2024 – Nov 2027

Skills

Languages: C, C++, Java, Python

Frameworks: Pandas, NumPy, Matplotlib, Seaborn, PyTorch, HuggingFace, OpenCV, Django, FastAPI, Spring Boot

Technologies: Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, Git/GitHub, Cloud Computing, LaTeX

References

- **Prof. Dr. Abdul Kadar Muhammad Masum**, Professor, Director of IQAC, & Dean, School of Engineering, Southeast University — Relationship: *Research Supervisor*
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- **Md. Mijanur Rahman**, Assistant Professor, Dept. of CSE, Southeast University — Relationship: *Undergraduate Thesis Supervisor*
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- **Dr. Md. Abul Kalam Azad**, Professor, Dept. of Business and Technology Management, Islamic University of Technology — Relationship: *Workplace Supervisor*
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